

Q1 - 24 June - Shift 1

A resistance of $40\ \Omega$ is connected to a source of alternating current rated 220 V , 50 Hz . Find the time taken by the current to change from its maximum value to rms value :

- (A) 2.5 ms (B) 1.25 ms
(C) 2.5 s (D) 0.25 s

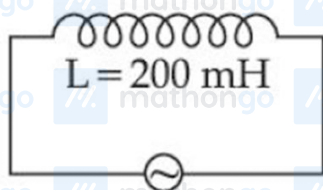
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Q2 - 24 June - Shift 1

As shown in the figure an inductor of inductance 200 mH is connected to an AC source of emf 220 V and frequency 50 Hz . The instantaneous voltage of the source is 0 V when the

peak value of current is $\frac{\sqrt{a}}{\pi}\text{ A}$. The value of a is

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Q3 - 24 June - Shift 2

Given below are two statements :

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Statement-I : The reactance of an ac circuit is zero. It is possible that the circuit contains a capacitor and an inductor.

Statement-II : In ac circuit, the average power delivered by the source never becomes zero.

In the light of the above statements, choose the correct answer from the options given below :

- (A) Both Statement I and Statement II are true.
- (B) Both Statement I and Statement II are false.
- (C) Statement I is true but Statement II is false.
- (D) Statement I is false but Statement II is true.

Q4 - 25 June - Shift 1

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Match List-I with List-II

List-I

List-II

(A) AC generator (I) Detects the presence of current in the circuit

(B) Galvanometer (II) Converts mechanical energy into electrical energy

(C) Transformer (III) Works on the principle of resonance in AC circuit

(D) Metal detector (IV) Changes an alternating voltage for smaller or greater value

Choose the **correct answer** from the options given below :-

(A) (A)–(II), B–(I), (C)–(IV), (D)–(III)

(B) (A)–(II), B–(I), (C)–(III), (D)–(IV)

(C) (A)–(III), B–(IV), (C)–(II), (D)–(I)

(D) (A)–(III), B–(I), (C)–(II), (D)–(IV)

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Q5 - 25 June - Shift 1

If wattless current flows in the AC circuit, then the circuit is

(A) Purely Resistive circuit

(B) Purely Inductive circuit

(C) LCR series circuit

(D) RC series circuit only

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Q6 - 25 June - Shift 2

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Questions

MathonGo

A sinusoidal voltage $V(t) = 210 \sin 3000t$ volt is applied to a series LCR circuit in which $L = 10$ mH, $C = 25 \mu\text{F}$ and $R = 100\Omega$. The phase difference (Φ) between the applied voltage and resultant current will be :

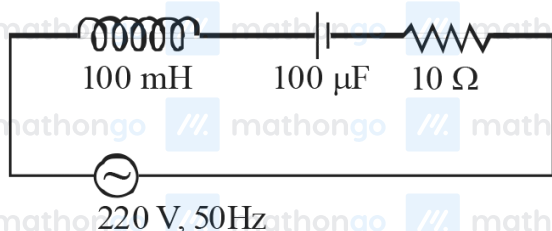
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- (A) $\tan^{-1}(0.17)$ (B) $\tan^{-1}(9.46)$
(C) $\tan^{-1}(0.30)$ (D) $\tan^{-1}(13.33)$

Q7 - 25 June - Shift 2

In a series LCR circuit, the inductance, capacitance and resistance are $L = 100\text{mH}$, $C = 100\mu\text{F}$ and $R = 10\Omega$ respectively. They are connected to an AC source of voltage 220V and frequency of 50 Hz . The approximate value of current in the circuit will be _____ A.

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Q8 - 26 June - Shift 1

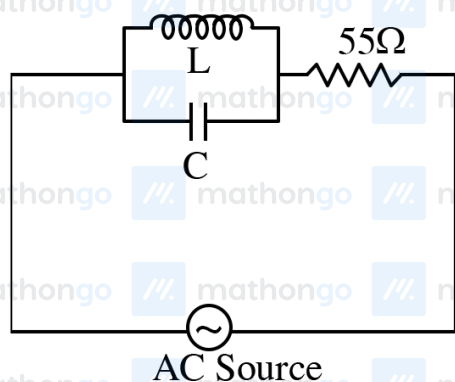
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Questions

MathonGo

A 110 V, 50 Hz, AC source is connected in the circuit (as shown in figure). The current through the resistance $55\ \Omega$, at resonance in the circuit, will be A.

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Q9 - 27 June - Shift 1

The current flowing through an ac circuit is given by

$$I = 5 \sin(120\pi t) \text{ A}$$

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How long will the current take to reach the peak value starting from zero?

- (A) $\frac{1}{60} \text{ s}$ (B) 60s
(C) $\frac{1}{120} \text{ s}$ (D) $\frac{1}{240} \text{ s}$

Q10 - 28 June - Shift 1

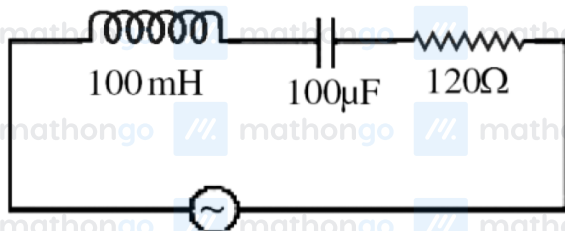
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Questions

MathonGo

An AC source is connected to an inductance of 100 mH, a capacitance of $100\ \mu\text{F}$ and a resistance of $120\ \Omega$ as shown in figure. The time in which the resistance having a thermal capacity $2\ \text{J}^\circ\text{C}$ will get heated by 16°C is _____ s.

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Q11 - 28 June - Shift 1

A telegraph line of length 100 km has a capacity of $0.01\ \mu\text{F}/\text{km}$ and it carries an alternating current at 0.5 kilo cycle per second. If minimum impedance is required, then the value of the inductance that needs to be introduced in series is _____ mH.

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(if $\pi = \sqrt{10}$)

Q12 - 28 June - Shift 2

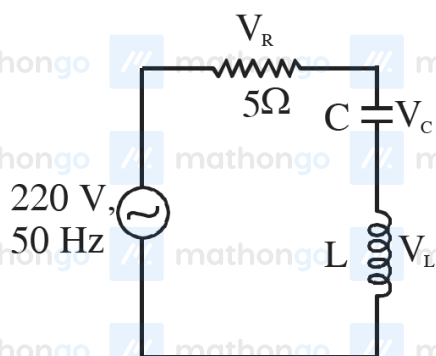
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Questions

MathonGo

In the given circuit, the magnitude of V_L and V_C are twice that of V_R . Given that $f = 50$ Hz, the inductance of the coil is $\frac{1}{K\pi}$ mH. The value of K is _____

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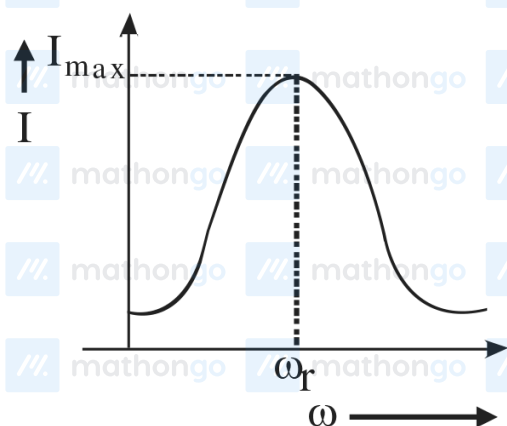
Q13 - 29 June - Shift 1

#MathBoleTohMathonGo

For a series LCR circuit, I vs ω curve is shown :

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- (a) To the left of ω_r , the circuit is mainly capacitive.
- (b) To the left of ω_r , the circuit is mainly inductive.
- (c) At ω_r , impedance of the circuit is equal to the resistance of the circuit.
- (d) At ω_r , impedance of the circuit is 0.



Choose the **most appropriate** answer from the options given below :

- (A) (a) and (d) only (B) (b) and (d) only
- (C) (a) and (c) only (D) (b) and (c) only

Q14 - 29 June - Shift 2

An inductor of 0.5 mH, a capacitor of 200 μ F and a resistor of 2 Ω are connected in series with a 220 V ac source. If the current is in phase with the emf, the frequency of ac source will be $___ \times 10^2$ Hz.

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Answer Key

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Q1 (A) Q2 (242) Q3 (C) Q4 (A)
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Q5 (B) Q6 (A) Q7 (22) Q8 (0)
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Q9 (D) Q10 (15) Q11 (100) Q12 (0)
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Q13 (C) Q14 (5)
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Q1 (A)

Considering sinusoidal AC.

$$\text{Phase at maximum value} = \frac{\pi}{2}$$

$$\text{Phase at rms value} = \frac{3\pi}{4}$$

$$\text{Thus phase change} = \frac{3\pi}{4} - \frac{\pi}{2} = \frac{\pi}{4}$$

$$\text{Now } \omega = 2\pi f$$

$$= 2\pi \times 50$$

$$= 100\pi$$

$$\text{time taken } t = \frac{\theta}{\omega} = \frac{\pi/4}{100\pi} = \frac{1}{400} \text{ s}$$

$$t = 2.5 \times 10^{-3} = 2.5 \text{ ms}$$

Q2 (242)

$$f = 50\text{Hz}$$

$$X_L = 2\pi fL$$

$$= 2\pi(50)(200 \times 10^{-3})$$

$$= 20\pi\Omega$$

$$i_0 = \frac{V_0}{X_L} \Rightarrow \frac{V_{\text{rms}} \sqrt{2}}{X_L}$$

$$= \frac{(220)\sqrt{2}}{20\pi} = \frac{11\sqrt{2}}{\pi}$$

$$i_0 = \frac{\sqrt{242}}{\pi}$$

Q3 (C)

if $R = 0$, $P = 0$

Q4 (A)

AC generator converts mechanical energy into electrical energy. Galvanometer shows deflection when current passes through it so it is used to show presence of current in any wire. Transformer is used to step up or step down the voltage. Metals detectors contain inductor coils and use principle of induction and resonance in AC circuit.

Q5 (B)

Purely Inductive circuit

$$\theta = \frac{\pi}{2}$$

$$\cos \frac{\pi}{2} = 0$$

Average power = 0

Q6 (A)

$$X_L = 10^{-2} \times 3000 = 30 \Omega$$

$$X_C = \frac{1}{3000 \times 25 \times 10^{-6}} = \frac{40}{3} \Omega$$

$$X = X_L - X_C$$

$$= 30 - \frac{40}{3} = \frac{50}{3}$$

$$\tan \delta = \frac{X}{R} = \frac{50}{3 \times 100} = \frac{1}{6}$$

$$\delta = \tan^{-1} \left(\frac{1}{6} \right) = \tan^{-1} (0.17)$$

Q7 (22)

$$X_L = \omega L = 2\pi \times 50 \times 10^{-1} = 10\pi$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi \times 50} \times 10^4 = \frac{100}{\pi}$$

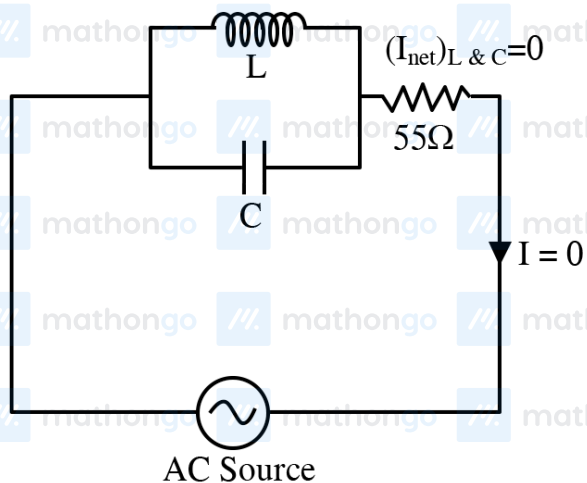
$$R = 10\Omega$$

$$Z = \sqrt{\left(10\pi - \frac{100}{\pi}\right)^2 + 10^2} \approx 10\Omega$$

$$i = \frac{E}{Z} \approx \frac{220}{10} \approx 22 \text{ Amp}$$

Q8 (0)

At resonance $I_L = I_C$



Alternatively,

$$\frac{1}{Z} = \sqrt{\left(\frac{1}{X_L} - \frac{1}{X_C} \right)^2}$$

At resonance, $X_L = X_C$ & $Z \rightarrow \infty$

$\therefore Z_{\text{total circuit}} \rightarrow \infty$ i.e, $I = 0$

Ans. 0

Q9 (D)

$$\omega = 120\pi = \frac{2\pi}{T} \Rightarrow T = \frac{1}{60} \text{ sec}$$

$$\text{time taken to reach peak value} = \frac{T}{4} = \frac{1}{240} \text{ s}$$

Q10 (15)

$$|(X_L - X_C)| = |10 - 10^2| = 90 \Omega$$

Z = Impedance

$$= \sqrt{(X_L - X_C)^2 + R^2} = \sqrt{(90)^2 + (20)^2} = 150 \Omega$$

$$i_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \left(\frac{2}{15} \right) \text{ A}$$

$$\text{Now } i_{\text{rms}}^2 R \Delta t = ms(\Delta T)$$

$$\Rightarrow \Delta t = 15 \text{ sec}$$

Q11 (100)

For minimum impedance

$$X_L = X_C$$

$$\Rightarrow \omega L = \frac{1}{\omega C} \Rightarrow L = \frac{1}{\omega^2 C} = 10^{-1} \text{ H} = 100 \text{ mH}$$

Q12 (0)

$$V_L = V_C = 2V_R$$

$$X_L = X_C = 2R$$

$$X_L = 10\Omega$$

$$\omega L = 10$$

$$2\pi fL = 10$$

$$L = \frac{10}{2\pi f} = \frac{1}{10\pi} \text{H} = \frac{1000}{10\pi} \text{mH}$$

$$L = \frac{1}{\frac{1}{100}\pi}; \quad K = \frac{1}{100} = 0.01 \approx 0$$

Q13 (C)

at ω_r , $X_C = X_L$

$$\Rightarrow \frac{1}{\omega_r C} = \omega_r L$$

So if $\omega < \omega_r$ then x_c will increase and X_L will decrease.

Hence to left of ω_r circuit is capacitive

$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

$$\text{at } \omega_r, Z = \sqrt{R^2 + 0^2} = R \text{ (C)}$$

Q14 (5)

If Current is in phase with emf then the frequency

$$\text{of source} = \frac{1}{2\pi\sqrt{LC}} \text{ (Resonant frequency)}$$

$$= \frac{1}{2\pi\sqrt{\frac{1}{2} \times 10^{-3} \times 2 \times 10^{-4}}}$$

$$= \frac{1}{2\pi} \times \sqrt{10} \times 1000 = 500 \text{ Hz}$$